

Lab Report 07

1. Objective

The objective of this lab is to understand the concept of one-dimensional arrays in C++, including declaration, initialization, accessing elements, copying arrays, and working with character arrays.

2. Apparatus / Software

- Computer System
 - C++ Compiler (Code::Blocks / Dev C++ / VS Code)
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3. Theory

An array is a collection of elements of the same data type stored in contiguous memory locations. Each element is accessed using an index starting from 0.

4. Lab Tasks Implementation

Task 1: Array Declaration and Initialization

```
#include <iostream>
using namespace std;

int main() {

    // integer array declare aur initialize ki ja rahi hai
    int numbers[5] = {10, 20, 30, 40, 50};

    // loop ke zariye array ke tamam elements display kiye ja rahe hain
    for (int i = 0; i < 5; i++) {
        cout << numbers[i] << endl;
    }

    return 0;
```

```
}
```

Task 2: Accessing Array Elements

```
#include <iostream>
using namespace std;

int main() {

    // character array declare ki ja rahi hai
    char message[] = "Hello";

    // loop ke zariye har character ko alag line par display kiya ja raha hai
    for (int i = 0; message[i] != '\0'; i++) {
        cout << message[i] << endl;
    }

    return 0;
}
```

Task 3: Copying Arrays

```
#include <iostream>
using namespace std;

int main() {

    // source aur destination arrays declare ki ja rahi hain
    int source[5] = {5, 10, 15, 20, 25};
    int destination[5];

    // loop ke zariye source array ko destination array me copy kiya ja raha
    hai
    for (int i = 0; i < 5; i++) {
        destination[i] = source[i];
    }

    // source array display ki ja rahi hai
    for (int i = 0; i < 5; i++) {
        cout << source[i] << " ";
    }

    cout << endl;

    // destination array display ki ja rahi hai
    for (int i = 0; i < 5; i++) {
        cout << destination[i] << " ";
    }

    cout << endl;

    return 0;
}
```

```
}
```

Task 4: Array Sum

```
#include <iostream>
using namespace std;

int main() {

    int arr[5];
    int sum = 0;

    // user se 5 numbers input liye ja rahe hain
    for (int i = 0; i < 5; i++) {
        cin >> arr[i];
    }

    // array ke tamam elements ka sum calculate kiya ja raha hai
    for (int i = 0; i < 5; i++) {
        sum = sum + arr[i];
    }

    // final sum display kiya ja raha hai
    cout << sum << endl;

    return 0;
}
```

5. Conclusion

In this lab, one-dimensional arrays were implemented successfully in C++. The tasks demonstrated declaration, initialization, element access, copying arrays, character array usage, and sum calculation using loops.